

DS Lab Cycle

Section 1: Basics

1. Write programs to demonstrate the use of storage classes (local variable, global variable, static variable, register variable) in C.
2. Use a menu-driven program to insert, search, delete and sort elements in an array using functions (use global variables)
3. Use a menu-driven program to insert, search, delete and sort elements in an array using functions (use only local variables)
4. Search for all the occurrences of an element in an integer array (positions)
5. Sort the array elements in ascending order (minimum three functions: read, disp and sort)
6. Display the array elements in the same order using a recursive function
7. Display array elements in reverse order using a recursive function .
8. Perform the following matrix operations
 - a. Adding two matrices
 - b. Subtraction of one matrix from another.
 - c. Multiplication of two matrices
 - d. Transpose of a matrix.
 - e. Determinant of a matrix(2x2 and 3x3)
9. Read a String and display it in the reverse order using the same array itself
10. Read n Strings and display them in the ascending order.

Section 2: Stack

11. Implement stack operations using arrays.
12. A letter means push and an asterisk means pop in the following sequence. Give the sequence of values returned by the pop operations when this sequence of operations is performed on an initially empty LIFO stack.

E A S * Y * Q U E * * * S T * * * I O * N * * *

13. Reverse a string using Stack
14. Convert an expression from infix expression to postfix using stack
15. Convert an expression from infix expression to prefix using stack
16. Evaluate an infix expression using stack

Session - 3 Polynomial using array

17. Read a polynomial and display
18. Add two polynomials
19. Subtract two polynomials
20. Multiply two polynomials

Session 4: Structure

21. Define a structure for dates with dd/mm/yyyy. Provide functions for reading, displaying and comparing two dates for equality
22. Define a structure for employees with eno,ename, esal and dno. Read n employees information and provide functions for the following:
 - a. Searching an employee by no
 - b. Sorting the employees by
 - i. Name
 - ii. Salary
 - c. Deleting an employee

Session 5: Polynomials using Structure

23. Read a polynomial and display it; use structure array
24. Add two polynomials
25. Subtract two polynomials
26. Multiply two polynomials

Session 6 - String

27. Read a String and display it in the reverse order using
 - a. Just print it in the reverse order
 - b. Reverse the string in the same array itself
 - c. Using stack - already completed in program 13.
28. Read n Strings and display them in the ascending order

Session-7

29. Demonstrate queue using array
30. Demonstrate circular queue using array.

Session 8: Dynamic Memory Allocation

31. Implement a) malloc , b) calloc and c) free functions
32. Use malloc to read n integers and find the mean.

33. Use calloc to read n numbers and find the mode.
34. Declare a structure for Books having author_name and book_name. Create an array of books using a pointer variable. Provide functions for reading n books and displaying the same using pointers.
35. Use realloc to implement varchar for any length.